Saurabh Rai

AI/ML Engineer | +91 930-574-4167 | rai359035@gmail.com | linkedin.com/in/Saurabh-rai | github.com/saurabhrai06

Education

Bennett University — Greater Noida, UP
B.Tech. in Computer Science and Engineering (Cloud Computing) — *September 2022 – September 2026*
CGPA: 8.4

Experience

Associate Product Intern — September 2025 – February 2026
goSTOPS — *Remote*

- Optimized two customer-facing AI chatbots for pre- and post-booking workflows by analyzing conversation data, identifying drop-offs and repeated queries, and improving self-service task completion.
- Collaborated with product, engineering, and support teams to improve intent handling, personalization, and customer resolution across booking journeys.

Software Development Intern — October 2024 – November 2024
SkillCraft Technology — *Remote*

- Contributed to the development, debugging, and optimization of 5+ software applications, improving system performance by 15% through code optimization and refactoring.
- Collaborated on 10+ feature releases and participated in 12+ code reviews, improving code quality, maintainability, and development workflows.

Projects

YouTube RAG Chatbot — Python, Streamlit, LangChain, Groq, FAISS, BM25 — September - 2026

- Engineered a RAG application using LangChain, Groq, FAISS, BM25, Sentence Transformers, semantic chunking, multilingual embeddings, MMR, and RRF-based hybrid retrieval to process YouTube transcripts.
- Designed query rewriting, multi-query retrieval, routing, contextual compression, context optimization, relevance and grounding guardrails, and timestamp citations for transcript-grounded answers.

Raven AI Chatbot — Python, Streamlit, LangGraph, Groq, SQLite — August - 2026

- Engineered a stateful LLM chatbot using Streamlit and LangGraph with graph-based conversation workflows, LLM response generation, UUID-based threads, and SQLite-backed persistence.
- Integrated conversation switching, message history, streaming responses, dynamic titles, structured REST APIs, and environment-based configuration for deployment.

AI Mock Interview — Next.js, React, Clerk, PostgreSQL, Gemini API — September - 2025

- Engineered an AI-driven mock interview platform using Gemini API to generate personalized interview experiences and feedback, with Clerk authentication and PostgreSQL persistence.
- Integrated full-stack interview workflows using Next.js, React, Drizzle ORM, ShadCN, and Tailwind CSS for interview management, response storage, and feedback delivery.

HelpNet – Civic Emergency Platform — HTML, CSS, JavaScript, Leaflet.js, Geolocation API — July - 2025

- Developing a real-time, location-aware civic assistance platform using Leaflet.js and Geolocation API for emergency reporting, incident visualization, and urgency-based filtering.
- Designed an offline-first architecture to preserve core functionality in low-connectivity environments, improving accessibility and resilience for underserved regions.

Research

Development-Based Classification of Satellite Imagery Using EfficientNetB0 — January 2025 – April 2025

- Engineered a TensorFlow/Keras image classification pipeline using EfficientNetB0 to classify 45 land-use categories from NWPU-RESISC45, achieving 84.4% accuracy.
- Leveraged transfer learning, fine-tuning, image augmentation, checkpointing, and early stopping, improving accuracy by over 10% compared with traditional CNN approaches.

Skills

AI/ML: Machine Learning, Deep Learning, NLP, Transformers, LLMs, RAG, Embeddings, Vector Search, Hybrid Retrieval, MMR, BM25, Semantic Chunking, Query Rewriting, Multi-Query Retrieval, Contextual Compression, Prompt Engineering, Guardrails, Computer Vision
Frameworks: LangChain, LangGraph, TensorFlow, Keras, Scikit-learn, Hugging Face, FastAPI, Streamlit, React, Next.js
Languages: Python, C++, JavaScript, TypeScript, SQL, HTML5, CSS3
Backend/Database: Node.js, Express.js, REST APIs, PostgreSQL, SQLite, Drizzle ORM
Tools: Git, GitHub, VS Code, Google Colab, JIRA, Notion, Dialogflow

Certifications

AWS Certified Cloud Practitioner – Amazon Web Services